Paired bootstrap | widened gamma

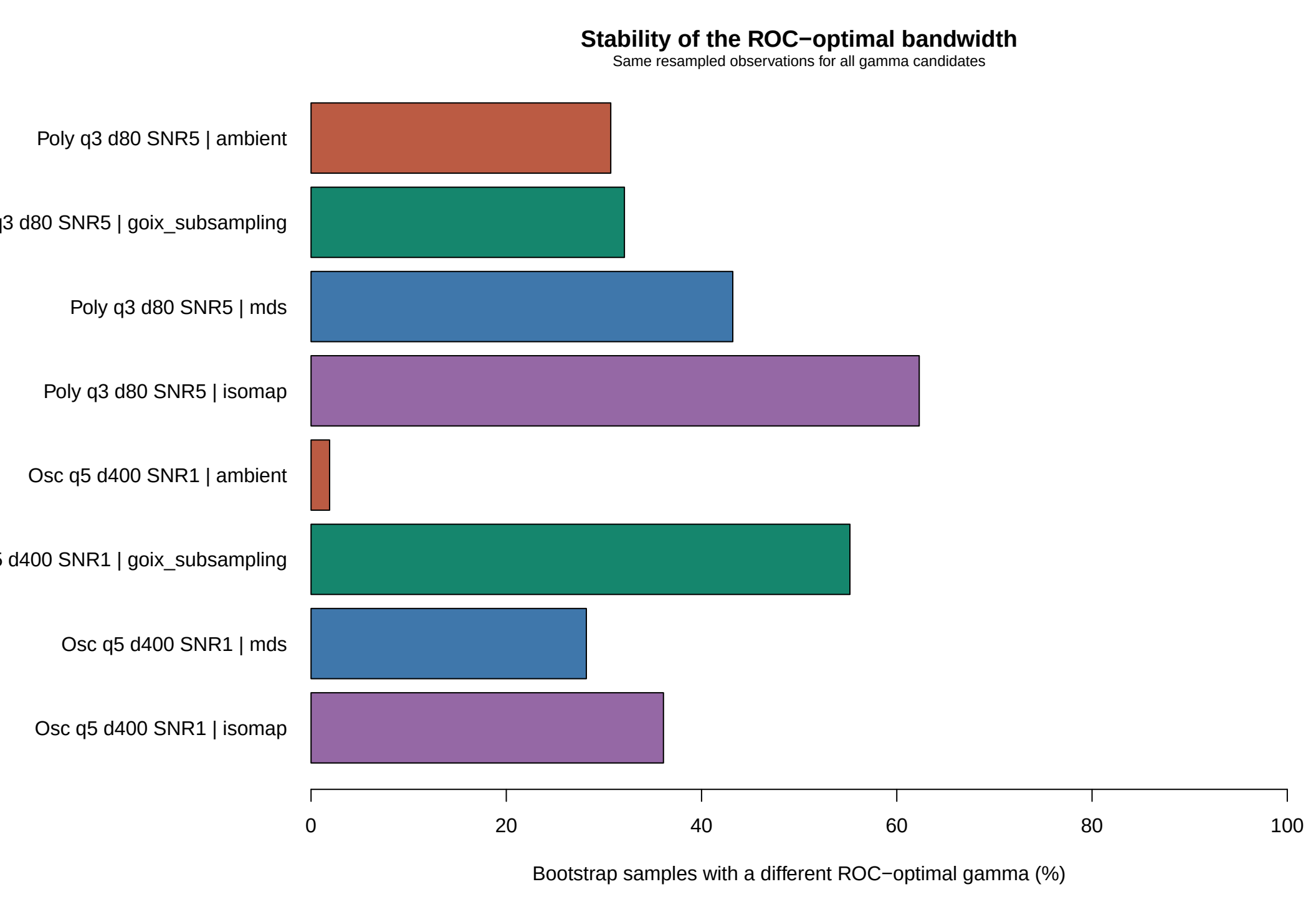
1000 paired, class−stratified resamples per synthetic case

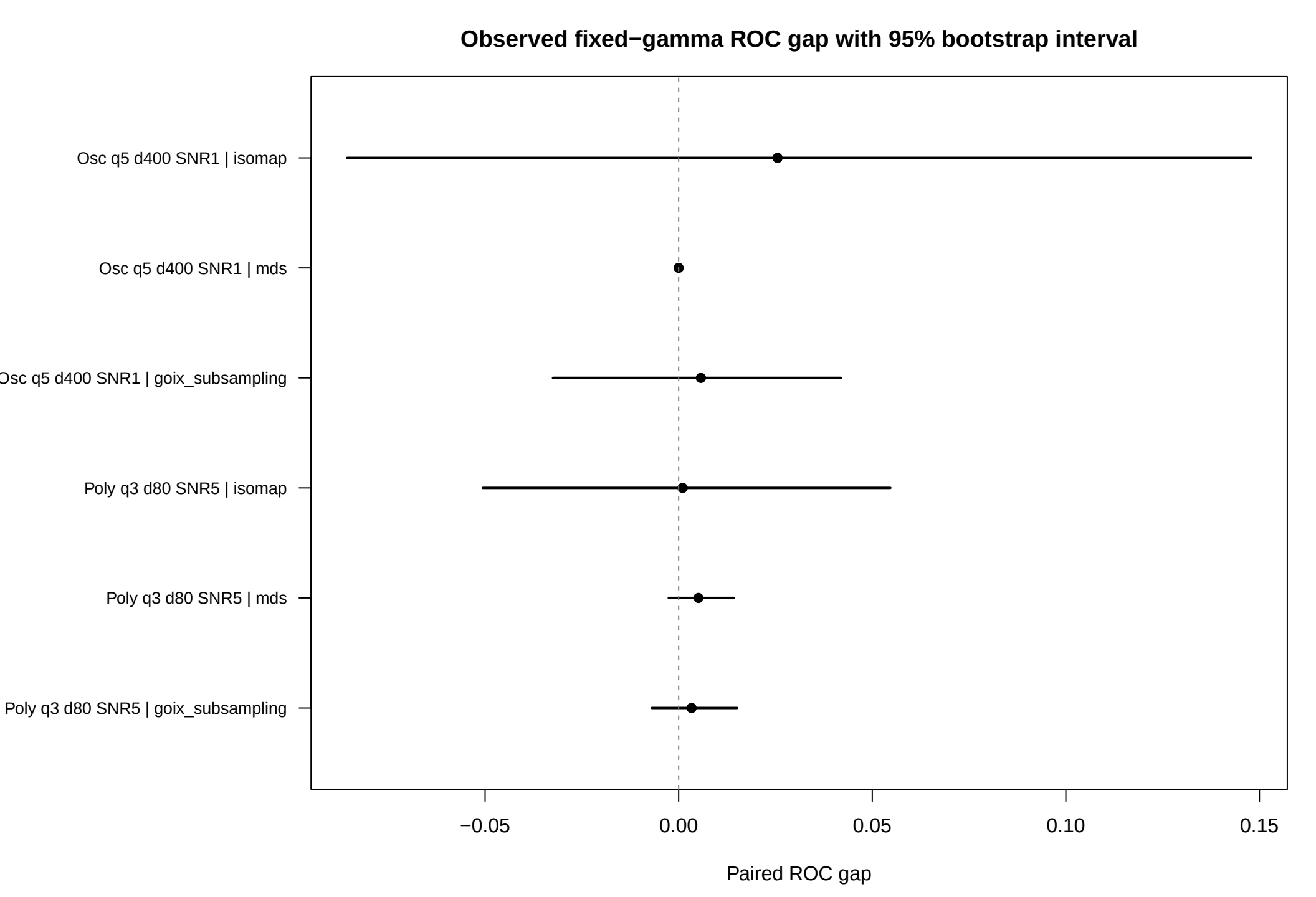
OCSVM fits and label scores held fixed; all gamma values share each resampled index set.

Case / representation	Anomalies	Selected	ROC best	Observed gap [95% CI]	Best changes
Poly q3 d80 SNR5 | ambient	22	−	64	not selected	30.7%
Poly q3 d80 SNR5 | goix_subsampling	22	4	64	0.003 [−0.007, 0.015]	32.1%
Poly q3 d80 SNR5 | mds	22	0.25	0.015625	0.005 [−0.003, 0.014]	43.2%
Poly q3 d80 SNR5 | isomap	22	1	4	0.001 [−0.051, 0.055]	62.3%
Osc q5 d400 SNR1 | ambient	22	−	4	not selected	1.9%
Osc q5 d400 SNR1 | goix_subsampling	22	1	64	0.006 [−0.032, 0.042]	55.2%
Osc q5 d400 SNR1 | mds	22	4	4	0.000 [0.000, 0.000]	28.2%
Osc q5 d400 SNR1 | isomap	22	4	64	0.026 [−0.086, 0.148]	36.1%

Gap = ROC at observed ROC−best gamma minus ROC at full−grid AUMVC choice.

No AUMVC choice implies missing gap, not zero regret. A tied ROC−best gamma is marked 'tie'.


Stability of the ROC−optimal bandwidth
Same resampled observations for all gamma candidates
Poly q3 d80 SNR5 | ambient
q3 d80 SNR5 | goix_subsampling
Poly q3 d80 SNR5 | mds
Poly q3 d80 SNR5 | isomap
Osc q5 d400 SNR1 | ambient
5 d400 SNR1 | goix_subsampling
Osc q5 d400 SNR1 | mds
Osc q5 d400 SNR1 | isomap
0
20
40
60
80
100
Bootstrap samples with a different ROC−optimal gamma (%)


Observed fixed−gamma ROC gap with 95% bootstrap interval
Osc q5 d400 SNR1 | isomap
Osc q5 d400 SNR1 | mds
Osc q5 d400 SNR1 | goix_subsampling
Poly q3 d80 SNR5 | isomap
Poly q3 d80 SNR5 | mds
Poly q3 d80 SNR5 | goix_subsampling
−0.05
0.00
0.05
0.10
0.15
Paired ROC gap